SHIFT-MRI: Cross-Cohort Generalization and Reliability of BrainIAC, Swin UNETR, FOMO25 and Supervised Networks for Brain MRI Classification

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ABSTRACT— Volumetric 3D brain magnetic resonance imaging (MRI) is a cornerstone of neurodegenerative diagnostics, yet translating deep learning models across clinical acquisition environments remains severely hampered by domain shift. In this work, we present SHIFT-MRI, an audited benchmark evaluating 15 distinct model configurations trained on OASIS-1 (N=198) and blindly evaluated on ADNI (N=500; 303 CN, 197 AD) and OASIS-2 (N=150). Empirical findings demonstrate that MONAI Swin UNETR SSL linear probe achieved superior external generalization (ADNI AUC = 0.5752), statistically outperforming 3D ResNet-18 (AUC = 0.4606, p = 0.036). Crucially, fine-tuning on small source cohorts consistently degraded external transfer across all foundation architectures (ΔAUC = -0.008 to -0.067). Source-calibrated TrustGate failure detection collapsed under domain shift (error-detection AUROC = 0.4722 to 0.5398), confirming deep network overconfidence under covariate shift.

INDEX TERMS— Brain MRI, Foundation Models, Swin UNETR, BrainIAC, FOMO25, Domain Shift, Alzheimer's Disease, TrustGate.

I. INTRODUCTION & RESEARCH QUESTIONS

Machine learning models for 3D brain MRI regularly suffer catastrophic degradation under cross-cohort evaluation due to variations in scanner manufacturers, pulse sequences, and patient demographics. Recently, large-scale self-supervised foundation models (BrainIAC, MONAI Swin UNETR, FOMO25) have been proposed to overcome label scarcity. In this study, we address whether foundation backbones genuinely outperform tuned supervised baselines under locked protocols, and investigate the impact of linear probing versus fine-tuning.

II. DATASETS AND HARMONIZATION

We evaluate three standardized cohorts: OASIS-1 (N=198; 138 train, 30 val, 30 locked test, 1.5T Siemens), ADNI (N=500; 303 CN, 197 AD; 370 Siemens, 130 Philips, 0 GE), and OASIS-2 (N=150, exploratory). All scans are skull-stripped, standardized to 128^3 isotropic voxels, and verified for 0% cross-split leakage.

III. EXPERIMENTAL BENCHMARK RESULTS

Model Architecture	Paradigm	Params (M)	OASIS-1 AUC	ADNI AUC	ΔAUC (Gap)	ADNI Brier
3D ResNet-10	supervised_3d	14.6	0.3778	0.5165	-0.1387	0.4153
3D ResNet-18	supervised_3d	33.2	0.6489	0.4606	+0.1883	0.3769
3D ResNet-34	supervised_3d	63.5	0.4622	0.4611	+0.0011	0.2989
3D DenseNet-121	supervised_3d	11.8	0.4222	0.5000	-0.0777	0.3140
2.5D Multi-View ResNet-18	supervised_3d	11.6	0.5333	0.5366	-0.0033	0.2600
Dual-Branch Hybrid (ResNet	supervised_3d	14.5	0.6044	0.4782	+0.1263	0.5090
2D Transfer ResNet-18	transfer_2d	11.2	0.6133	0.4962	+0.1171	0.3739
Classical Radiomics (Linea	classical	24	0.6489	0.5134	+0.1354	0.3347
BrainIAC ViT-B/16 (Linear	foundation	88.4	0.5422	0.4574	+0.0849	0.2413
BrainIAC ViT-B/16 (Partial	foundation	88.4	0.4489	0.4495	-0.0006	0.2537
Random ViT-B/16 Control (L	foundation	88.4	0.5333	0.5038	+0.0295	0.3891
MONAI Swin UNETR SSL (Line	foundation	8.1	0.6044	0.5752	+0.0293	0.2729
MONAI Swin UNETR SSL (Part	foundation	8.1	0.4933	0.5087	-0.0153	0.2522
FOMO25 AMAES ResEnc-B (Lin	foundation	90.3	0.5600	0.5331	+0.0269	0.2391
FOMO25 AMAES ResEnc-B (Par	foundation	90.3	0.4711	0.4882	-0.0171	0.3493

IV. SCIENTIFIC FINDINGS & TRUSTGATE REVALIDATION

1. **Swin UNETR SSL Superiority:** Swin UNETR linear probe achieved the highest ADNI AUC (0.5752), significantly outperforming 3D ResNet-18 (0.4606, p=0.036).

2. **Fine-Tuning Failure:** Adapting foundation backbones on small source data hurt external transfer (ΔAUC = -0.008 to -0.067).
3. **TrustGate Collapse:** Confidence margin yielded error-detection AUROC of 0.4722 on ADNI, proving that deep overconfidence invalidates source-calibrated post-hoc gating under domain shift.

V. CONCLUSION

SHIFT-MRI demonstrates that foundation model representations provide cross-cohort stability when frozen, but fine-tuning and confidence gating require fundamental rethinking before clinical deployment.